



Lessons

- Table of Contents
- Confidence Levels
- Notes
- Bookmarks
- Highlights

Adding and Subtracting...

< >

Adding and Subtracting Polynomials

Toolbox

Adding and Subtracting Polynomials

All algebraic expressions are made up of **terms**. A term can be a variable, a number (also called a **constant**), or one or more variables multiplied by a number.

You have already worked with terms. Here are some examples:

x

$3y$

-5

a^2b

$\frac{w}{8}$

Like terms have exactly the same variables and exponents. The terms x^3 and $4x^3$ are like terms because they have the same variable raised to the same power. **Unlike terms** have different variables or the same variable raised to different powers.

When a variable is multiplied by a number, the number is called a **coefficient**. The coefficient of $5ab$ is 5. Like terms can have different coefficients.

A **polynomial** is an expression with one term or the sum or difference of unlike terms. Polynomials are classified by the number of terms they contain. A **monomial** has one term. The examples at the top of the page are all monomials.

A **binomial** is the sum or difference of two unlike terms. The following are binomials:

$x + 8y$

$6y^2 - 1$

$a^3 + 5ab$

A **trinomial** is the sum or difference of three unlike terms. The following are trinomials:

$x^2 + 5x + 6$

$9x^3 - 7y + z^2$

$12a^2 - 17ab - 7b^2$

To simplify polynomials, combine like terms.

Example 1

Hide

Simplify: $5x^2 - 3x + 4 + 6x - 9$

Step 1

Group like terms.

$$5x^2 - 3x + 6x + 4 - 9$$

Step 2

Combine like terms by adding the coefficients.

$$5x^2 - 3x + 6x + 4 - 9 = 5x^2 + 3x - 5$$

Step 3

Write the simplified expression.

$$5x^2 + 3x - 5$$

When you subtract polynomials, you will need to distribute the minus sign to the terms in parentheses that follow.

Example 2

Hide

Simplify: $(y^2 - 2y) - (2y^2 + 5y)$

Step 1

Use the distributive property.

$$(y^2 - 2y) - (2y^2 + 5y)$$

$$= y^2 - 2y - 2y^2 - 5y$$

Step 2

Group like terms.

$$y^2 - 2y^2 - 2y - 5y$$

Step 3

Combine like terms.

$$-y^2 - 7y$$

Key Terms

 **binomial**

 **constant**

 **monomial**

 **term**

 **coefficient**

 **like terms**

 **polynomial**

 **unlike terms**

Rate Your Confidence

 High

 Medium

 Low

Continue >>

Category

Lesson 1: Polynomial Operations

 **Related Questions:**
[Practice questions taken related to this topic](#)

 Report Content Errors